

Sharp Growth for Absolute-Loss Max-Linear Regression: A Deterministic Polyhedral Error Bound

Solution note

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Abstract

We prove a deterministic sharp-growth/error-bound theorem for the absolute-loss max-linear regression objective

$$f(\beta_1, \dots, \beta_r) = \frac{1}{m} \sum_{i=1}^m \left| \bar{y}_i - \max_{j \in [r]} \langle \beta_j, a_i \rangle \right|.$$

Assuming only that the global minimizer set is nonempty, we show that there are constants $\epsilon > 0$ and $\mu > 0$ such that

$$f(\beta) - f^* \geq \mu \operatorname{dist}(\beta, X^*) \quad \text{whenever } \operatorname{dist}(\beta, X^*) < \epsilon.$$

The proof is finite-dimensional and polyhedral: the objective is continuous piecewise affine, its active-index/residual-sign regions are polyhedra, and Hoffman's error bound gives a uniform linear growth estimate on every solution-adjacent cell.

Status relative to the Charisopoulos–Davis problem. This note proves the sharp-growth assumption (A1) for the stated max-linear absolute-loss objective. Thus it is a solution to the A1 sharp-growth part of the problem. It does not prove the separate semismoothness/(b)-regularity assumption (A2) in the non-isolated-solution regime; hence, if the target is the simultaneous verification of the full A1+A2 hypothesis package, the note should be viewed as partial progress.

1 Problem statement and result

Let $a_1, \dots, a_m \in \mathbb{R}^d$, $\bar{y} \in \mathbb{R}^m$, and $r \geq 1$. Put

$$\mathcal{E} := (\mathbb{R}^d)^r, \quad \beta = (\beta_1, \dots, \beta_r) \in \mathcal{E},$$

and equip $\mathcal{E}$ with its Euclidean norm. For $i \in [m]$ and $j \in [r]$, define the linear forms

$$u_{ij}(\beta) := \langle \beta_j, a_i \rangle,$$

the max-linear prediction

$$\hat{y}_i(\beta) := \max_{j \in [r]} u_{ij}(\beta),$$

and the absolute-loss objective

$$f(\beta) := \frac{1}{m} \sum_{i=1}^m |\bar{y}_i - \hat{y}_i(\beta)|. \tag{1}$$

Let

$$f^* := \inf_{\beta \in \mathcal{E}} f(\beta), \quad X^* := \operatorname{argmin}_{\beta \in \mathcal{E}} f(\beta),$$

and assume throughout that $X^* \neq \emptyset$. Since $f \geq 0$, this also implies that f^* is finite.

The sharp-growth assumption used by Charisopoulos and Davis is an error bound of the form

$$f(\beta) - f^* \geq \mu \operatorname{dist}(\beta, X^*) \quad (2)$$

near the solution set, or near a fixed solution. We prove the following stronger tube version.

Theorem 1.1 (Uniform tube sharp growth). *For the objective (1), if $X^* \neq \emptyset$, then there exist constants $\epsilon > 0$ and $\mu > 0$ such that*

$$f(\beta) - f^* \geq \mu \operatorname{dist}(\beta, X^*) \quad \text{for every } \beta \in \mathcal{E} \text{ with } \operatorname{dist}(\beta, X^*) < \epsilon. \quad (3)$$

Consequently, for every $\bar{\beta} \in X^$, there is a neighborhood U of $\bar{\beta}$ such that (2) holds on U .*

Corollary 1.2 (Noiseless statistical model). *Suppose the observations are generated by a max-linear model*

$$\bar{y}_i = \max_{j \in [r]} \langle \bar{\beta}_j, a_i \rangle, \quad i = 1, \dots, m,$$

for some tuple $\bar{\beta} = (\bar{\beta}_1, \dots, \bar{\beta}_r) \in \mathcal{E}$. Then $f^ = 0$ and $X^* \neq \emptyset$, and therefore the sharp-growth estimate (3) holds deterministically for that realization of the data. In particular, it holds almost surely under any random design model, including the Gaussian design model, whenever the observations are noiselessly generated in this way.*

2 The active-index and residual-sign decomposition

The proof begins by writing f as a continuous piecewise affine function on an explicit finite polyhedral cover.

For a vector of active max-indices $j = (j_1, \dots, j_m) \in [r]^m$ and a residual-sign vector $\sigma = (\sigma_1, \dots, \sigma_m) \in \{-1, +1\}^m$, define the closed polyhedron

$$P_{j,\sigma} := \left\{ \beta \in \mathcal{E} : \begin{aligned} &u_{ij_i}(\beta) \geq u_{ik}(\beta) \quad \forall i \in [m], \forall k \in [r], \\ &\sigma_i(\bar{y}_i - u_{ij_i}(\beta)) \geq 0 \quad \forall i \in [m] \end{aligned} \right\}. \quad (4)$$

If $\beta \in P_{j,\sigma}$, then j_i is an active maximizer for the i -th sample and the residual $\bar{y}_i - u_{ij_i}(\beta)$ has sign σ_i , with either sign allowed when the residual is zero. Hence, on $P_{j,\sigma}$,

$$f(\beta) = L_{j,\sigma}(\beta) := \frac{1}{m} \sum_{i=1}^m \sigma_i(\bar{y}_i - u_{ij_i}(\beta)), \quad (5)$$

where $L_{j,\sigma}$ is affine.

The cells $P_{j,\sigma}$ cover $\mathcal{E}$. Indeed, given β , choose for every i an index $j_i \in \operatorname{argmax}_{k \in [r]} u_{ik}(\beta)$, and choose $\sigma_i \in \{-1, +1\}$ so that $\sigma_i(\bar{y}_i - u_{ij_i}(\beta)) \geq 0$. If the residual is zero, either sign is valid.

Let

$$\mathcal{T} := [r]^m \times \{-1, +1\}^m$$

be the finite cell-index set. For $\tau = (j, \sigma) \in \mathcal{T}$, write

$$P_\tau := P_{j,\sigma}, \quad L_\tau := L_{j,\sigma}, \quad S_\tau := P_\tau \cap X^*.$$

On every cell,

$$S_\tau = \{\beta \in P_\tau : L_\tau(\beta) = f^*\}. \quad (6)$$

Thus, whenever nonempty, S_τ is a polyhedron. Also,

$$X^* = \bigcup_{\tau \in \mathcal{T}} S_\tau, \quad (7)$$

because the cells cover $\mathcal{E}$.

3 Two elementary polyhedral facts

We use two standard consequences of polyhedrality. The first is Hoffman's error bound, specialized to a face exposed by a single affine function.

Lemma 3.1 (Hoffman bound for an affine face). *Let $P \subset \mathbb{R}^n$ be a nonempty polyhedron and let $\ell : \mathbb{R}^n \rightarrow \mathbb{R}$ be affine. Suppose*

$$\ell(x) \geq 0 \quad \forall x \in P, \quad F := \{x \in P : \ell(x) = 0\} \neq \emptyset.$$

Then there is a constant $H < \infty$ such that

$$\text{dist}(x, F) \leq H \ell(x) \quad \forall x \in P. \quad (8)$$

If $\ell \equiv 0$ on P , then $F = P$ and the conclusion is trivial.

Proof. Write $P = \{x : Ax \leq b\}$. The set F is the solution set of the finite linear system

$$Ax \leq b, \quad \ell(x) = 0.$$

Hoffman's lemma for finite systems of linear inequalities and equalities gives a constant $H < \infty$ such that

$$\text{dist}(x, F) \leq H(\|(Ax - b)_+\| + |\ell(x)|) \quad \forall x \in \mathbb{R}^n.$$

For $x \in P$, the inequality residual vanishes. Since $\ell(x) \geq 0$ on P , the displayed estimate reduces to (8). $\square$

Lemma 3.2 (Separation of disjoint polyhedra). *If $P, Q \subset \mathbb{R}^n$ are nonempty polyhedra and $P \cap Q = \emptyset$, then*

$$\text{dist}(P, Q) > 0.$$

Consequently, if $Q_1, \dots, Q_N$ are nonempty polyhedra and P is a polyhedron disjoint from $\bigcup_{k=1}^N Q_k$, then

$$\text{dist}\left(P, \bigcup_{k=1}^N Q_k\right) > 0.$$

Proof. The Minkowski difference

$$P - Q := \{p - q : p \in P, q \in Q\}$$

is a polyhedron: it is the linear image, equivalently the projection, of the polyhedron

$$\{(p, q, z) : p \in P, q \in Q, z = p - q\}.$$

Therefore $P - Q$ is closed. If $\text{dist}(P, Q) = 0$, then $0 \in \text{cl}(P - Q) = P - Q$, so $p - q = 0$ for some $p \in P$ and $q \in Q$. This gives $p = q \in P \cap Q$, a contradiction. Hence $\text{dist}(P, Q) > 0$.

For the finite-union statement, apply the first part to each pair (P, Q_k) and then take the minimum over finitely many positive distances. $\square$

4 Proof of Theorem 1.1

We now prove the tube error bound.

For each cell P_τ with $S_\tau \neq \emptyset$, define

$$\ell_\tau(\beta) := L_\tau(\beta) - f^*.$$

Since $f = L_\tau$ on P_τ and $f \geq f^*$, one has $\ell_\tau \geq 0$ on P_τ . Moreover,

$$S_\tau = \{\beta \in P_\tau : \ell_\tau(\beta) = 0\}.$$

If $\ell_\tau \equiv 0$ on P_τ , then $P_\tau \subset X^*$. If $\ell_\tau \not\equiv 0$ on P_τ , Lemma 3.1 gives a finite constant H_τ such that

$$\text{dist}(\beta, S_\tau) \leq H_\tau (L_\tau(\beta) - f^*) \quad \forall \beta \in P_\tau. \quad (9)$$

Because $\mathcal{T}$ is finite, the number

$$\mu := \begin{cases} \min\{H_\tau^{-1} : S_\tau \neq \emptyset, \ell_\tau \not\equiv 0 \text{ on } P_\tau\}, & \text{if this index set is nonempty,} \\ 1, & \text{otherwise} \end{cases} \quad (10)$$

is strictly positive.

It remains only to rule out cells that do not touch the solution set. Let

$$\mathcal{B} := \{\tau \in \mathcal{T} : S_\tau = \emptyset\}$$

be the set of bad cells. If $\tau \in \mathcal{B}$ and $P_\tau \neq \emptyset$, then $P_\tau \cap X^* = \emptyset$. By (7), the set X^* is a finite union of the nonempty polyhedra S_ν . Lemma 3.2 therefore gives

$$\text{dist}(P_\tau, X^*) > 0.$$

Empty cells may be ignored. Choose $\epsilon > 0$ so that

$$0 < \epsilon < \text{dist}(P_\tau, X^*) \quad \text{for every nonempty bad cell } P_\tau. \quad (11)$$

For example, if at least one nonempty bad cell exists, take one half of the minimum of these finitely many positive distances; if no nonempty bad cell exists, take any $\epsilon > 0$.

Now let $\beta \in \mathcal{E}$ satisfy $\text{dist}(\beta, X^*) < \epsilon$. Choose a cell P_τ containing β . By (11), this cell cannot be bad, and hence $S_\tau \neq \emptyset$. If $\ell_\tau \equiv 0$ on P_τ , then $P_\tau \subset X^*$, so $\beta \in X^*$ and the desired inequality is immediate. Otherwise, using (5), (9), (10), and $S_\tau \subset X^*$, we get

$$f(\beta) - f^* = L_\tau(\beta) - f^* \geq \mu \text{dist}(\beta, S_\tau) \geq \mu \text{dist}(\beta, X^*).$$

This proves (3).

Finally, fix any $\bar{\beta} \in X^*$. Since

$$\text{dist}(\beta, X^*) \leq \|\beta - \bar{\beta}\|,$$

the ball $U := \{\beta : \|\beta - \bar{\beta}\| < \epsilon\}$ is contained in the tube $\{\beta : \text{dist}(\beta, X^*) < \epsilon\}$. Therefore (2) holds throughout U . This completes the proof.

5 Interpretation and limitations

The proof shows that sharp growth for (1) is not a Gaussian or generic-position phenomenon. It is a deterministic consequence of finite polyhedrality. The active-index/residual-sign cells are finitely many polyhedra; on each cell the objective is affine; and near the solution set, cells not meeting the solution set are separated away. On every cell that does meet X^* , Hoffman’s bound converts the affine optimality gap into distance to the optimal face, and hence into distance to X^* .

The constants μ and ϵ are explicit in principle. They are determined by finitely many linear systems consisting of the max-activity inequalities, the residual-sign inequalities, and the affine equations $L_\tau = f^*$. This note does not give a convenient statistical lower bound on μ in terms of (m, d, r) or the distribution of the measurements; it proves existence for every fixed instance with $X^* \neq \emptyset$.

Remark 5.1 (No identifiability claim). The theorem proves sharp growth with respect to the full minimizer set X^* , exactly as required in (2). It does not assert identifiability of a particular ground-truth tuple $(\bar{\beta}_1, \dots, \bar{\beta}_r)$. Such identifiability may fail because of permutations, repeated components, or insufficient measurements. None of these failures affects the error bound above, since the right-hand side uses $\text{dist}(\beta, X^*)$, not distance to one selected parameter tuple.

Remark 5.2 (Relation to A2). This note proves the sharp-growth assumption (A1). It does not prove the separate semismoothness or (b) -regularity assumption (A2) needed in the full convergence theory of Charisopoulos and Davis. In the isolated-solution case, the paper notes that semialgebraicity can be used to obtain A2; the non-isolated A2 verification is outside the scope of this note.

References

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